Grade 11 – Term 1
C1–C7 Catch-up Exam Version 2 Solutions
Continuous distributions and sample data

Global Economics – 2026–2027

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Learning objective

Identify probability distributions and their supports; use geometric interval probabilities and exponential waiting times; calculate theoretical moments, empirical probabilities, and sample statistics; and interpret errors relative to theoretical models in economic and operational contexts.

Evaluated criteria

C1. Characterises discrete and continuous random variables, identifies suitable distributions and supports, and interprets density and cumulative probability in context.

C2. Applies uniform and triangular models and constant or piecewise-linear densities to interval probabilities using geometric areas.

C3. Uses circular geometry and arc proportions to calculate probabilities for uniform timing and directions around a circle.

C4. Applies exponential models to constant-rate waiting times, infers event rates, and assesses model suitability.

C5. Calculates and interprets the mean, variance, and standard deviation of continuous distributions using standard formulas.

C6. Estimates interval probabilities using sample relative frequencies and compares them with theoretical probabilities.

C7. Calculates sample means, variances, and standard deviations for different sample sizes and data representations and compares them with theory.

Assessment information

Each task shows its points. Passing requires **at least 5 cumulative points per criterion, including previous learning activities**. Show all procedures; answers without the required procedure receive no credit. Partial credit may be awarded for a correct method with arithmetic errors. Missing names or late submissions incur a 1-point deduction per criterion.

Criterion	C1	C2	C3	C4	C5	C6	C7	Total
Points	6	6	6	7	6	6	6	43

Conventions. $N(\mu, \sigma^2)$ uses mean and variance; $\text{Tri}(a, b, c)$ means minimum, maximum and mode; $\text{Exp}(\lambda)$ uses the rate parameter. Supports include their finite boundary points; endpoint inclusion does not affect continuous probabilities. Probabilities are rounded to four decimal places (percentages to two), statistics, rates and times to three where needed, and percentage errors to two. All errors use unrounded results. Sample variance uses $n - 1$. Blue task text links to its solution; each solution links back to its task. The arrows return to the problem and contents.

1 Models in Colombian operations [C1-6]

For each variable below, classify it as discrete or continuous, identify its distribution, parameters and support, and briefly justify your choice.

Questions/Tasks

C1 1. Municipal compost participation. A programme records (a) whether a randomly selected household sorts organic waste correctly (1 if correct, 0 otherwise); and (b) the number sorting correctly among 35 households. Households sort correctly independently with probability 0.86. **[1]**

C1 2. Water-meter auditing. A utility records (a) the number of independent tests up to and including the first faulty meter, with constant fault probability 0.06; and (b) a calibration error on $[-2, 4]$ litres, with equal probabilities for equal-length intervals and no errors outside this range. Give (a) the probability of the first fault on test k and (b) the density function. **[1]**

C1 3. Cultural-centre bookings. A centre records (a) the number of rehearsal bookings in one hour: arrivals are independent at a constant rate of 3 per hour, one at a time, with negligible multiple arrivals in sufficiently short intervals; and (b) room-reset time on $[8, 26]$ minutes: its density rises linearly from zero at 8 to a single peak at 15, then falls linearly to zero at 26, and is zero elsewhere. **[1]**

C1 4. Solar-energy monitoring. A cooperative records (a) the usable battery-bank fraction at sunset on $[0.2, 0.9]$: its density rises linearly from zero at 0.2 to its maximum at 0.9 and is zero elsewhere; and (b) sensor error with an exact symmetric bell curve on the real line, centred at 1 watt with standard deviation 3 watts. For (a), find the normalised density function and maximum height. **[1]**

C1 5. Agricultural training. A service records (a) the number of cancellations among 45 independently booked places, each cancelled with probability 0.07; and (b) the number of registration calls over four days: arrivals are independent at a constant rate of 6 per day, one at a time, with negligible multiple arrivals in sufficiently short intervals. **[1]**

C1 6. Community-kitchen planning. A kitchen studies (a) vegetable-batch preparation time from 12 to 36 minutes, with short times most common and density decreasing steadily to zero at the upper limit; and (b) daily rice use, symmetric around 18 kg, with values farther from 18 less common and standard deviation 2.5 kg. For each, discuss whether the distribution and support are uniquely determined. Propose a model, state any additional assumptions, and distinguish model support from possible physical values. Give the density function for (a) under your assumption. **[1]**

Solution: Compost participation

(a) Discrete: $X \sim \text{Bernoulli}(0.86)$, support $\{0, 1\}$. One household gives a single correct/incorrect outcome. **(b)** Discrete: $Y \sim \text{Bin}(35, 0.86)$, support $\{0, 1, \ldots, 35\}$. This counts successes in 35 independent trials with the same probability.

Return to this task

Solution: Water-meter auditing

(a) Discrete: $X \sim \text{Geom}(0.06)$, support $\{1, 2, \ldots\}$. Count every test up to and including the first faulty meter; $P(X = k) = 0.94^{k-1}(0.06)$. **(b)** Continuous: $Y \sim U(-2, 4)$ litres, support $[-2, 4]$ L. Equal-length intervals have equal probabilities, so $f(y) = 1/(4 - (-2)) = 1/6$ L^{-1} on the support and zero elsewhere.

Return to this task

Solution: Cultural-centre bookings

(a) Discrete: $X \sim \text{Poisson}(3)$, support $\{0, 1, 2, \ldots\}$. The one-hour mean is $3 \times 1 = 3$ bookings; the stated conditions define a Poisson process. **(b)** Continuous: $Y \sim \text{Tri}(8, 26, 15)$ minutes, support $[8, 26]$ min. The straight branches specify a triangular density with mode 15.

Return to this task

Solution: Solar-energy monitoring

(a) Continuous: increasing-linear density, equivalently $X \sim \text{Tri}(0.2, 0.9, 0.9)$, support $[0.2, 0.9]$. Normalisation gives height $2/0.7 = 20/7$ and $f(x) = 2(x - 0.2)/(0.7)^2 = 200(x - 0.2)/49$ on this support, zero elsewhere. The fraction is unitless; its mode is 0.9. **(b)** Continuous: $Y \sim N(1, 9)$ in watts, support $\mathbb{R}$. The exact symmetric bell curve identifies normality, with SD 3 W and variance $3^2 = 9$ W^2.

Return to this task

Solution: Agricultural training

(a) Discrete: $X \sim \text{Bin}(45, 0.07)$, support $\{0, 1, \ldots, 45\}$. These are 45 independent cancellation trials with constant probability. **(b)** Discrete: $Y \sim \text{Poisson}(24)$, support $\{0, 1, 2, \ldots\}$. The four-day mean is $6 \times 4 = 24$ calls; 6 is a daily rate, not the four-day count parameter.

Return to this task

Solution: Community-kitchen planning

(a) Continuous, support $[12, 36]$ min. "Decreases steadily" does not uniquely establish a straight-line density. *If a decreasing straight line is intended*, use $X \sim \text{Tri}(12, 36, 12)$ with $f(x) = (36 - x)/288$ on this support and zero elsewhere. Its height is $2/24 = 1/12$ min^{-1}; no extra parameter is needed under this assumption. **(b)** Continuous rice use. Symmetric concentration near 18 kg and SD 2.5 kg do not uniquely determine normality or full support. *If normality is assumed*, $Y \sim N(18, 6.25)$ in kg, model support $\mathbb{R}$, with variance $2.5^2 = 6.25$ kg^2. Physical rice use is nonnegative, but its exact physical support is not supplied. These are candidate models under explicit assumptions.

Return to this task

2 Seedling-tray preparation [C2-6]

At a Boyacá nursery, tray-filling and seeding time S has support $[4, 28]$ minutes. Its density rises linearly from zero at 4 to a single maximum at 14, then falls linearly to zero at 28; it is zero elsewhere.

Questions/Tasks

C2 Identify the model and support, sketch the density, find its maximum height and full piecewise formula (including outside the support), and calculate $P(8 \le S \le 22)$ using geometric areas. **[3]**

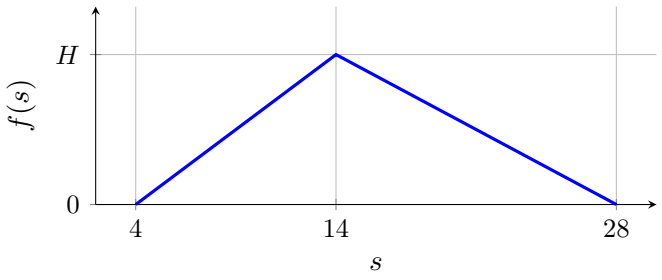
C2 Change the support to $[2, 34]$ minutes, keeping zero density at both endpoints and linear sides meeting at the mode of 14 minutes. Identify the new model, sketch its density, find its maximum height and full piecewise formula, and recalculate $P(8 \le S \le 22)$ using geometric areas. **[3]**

Solution: Original range

Stage 1: Support and shape

$S \sim \text{Tri}(4, 28, 14)$ has support $[4, 28]$ min. The density rises from $(4, 0)$ to $(14, H)$ and falls to $(28, 0)$.

Support and triangular shape; horizontal coordinate in minutes



Stage 2: Total area = 1

The triangle area is $\frac{1}{2} \times$ base $\times$ height, so

$$1 = \frac{1}{2}(28 - 4)H \quad \Longrightarrow \quad H = \frac{2}{24} = \frac{1}{12} \text{ min}^{-1}.$$

Stage 3: Complete piecewise pdf

Use $m = (y_2 - y_1)/(x_2 - x_1)$ and $q = y_1 - mx_1$ in $f(s) = ms + q$. For the rising line through $(4, 0)$ and $(14, \frac{1}{12})$,

$$m_1 = \frac{\frac{1}{12} - 0}{14 - 4} = \frac{1}{120}, \qquad q_1 = 0 - (\frac{1}{120})(4) = \frac{-1}{30}, \qquad f_1(s) = \frac{s - 4}{120}.$$

For the falling line through $(14, \frac{1}{12})$ and $(28, 0)$,

$$m_2 = \frac{0 - \frac{1}{12}}{28 - 14} = \frac{-1}{168}, \qquad q_2 = 0 - (\frac{-1}{168})(28) = \frac{1}{6}.$$

Thus the complete density is

$$f(s) = \begin{cases} \dfrac{s-4}{120}, & 4 \le s \le 14, \\ \dfrac{28-s}{168}, & 14 < s \le 28, \\ 0, & \text{otherwise.} \end{cases}$$

The branches meet at height $\frac{1}{12}$, are nonnegative, and enclose total area 1.

Stage 4: Probability by complementary tail triangles

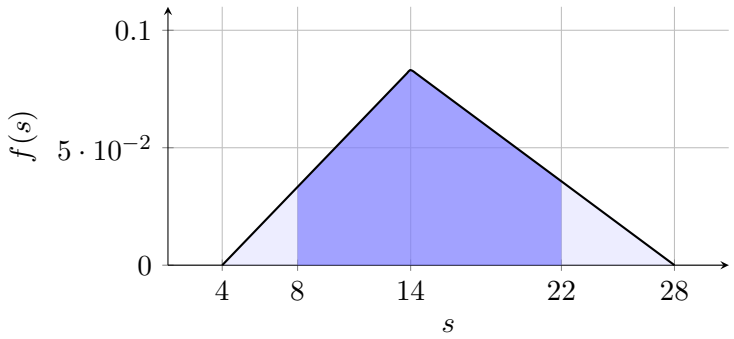
The interval straddles the mode; both tails must be removed:

$$P(8 \le S \le 22) = 1 - \frac{1}{2}(8-4)f(8) - \frac{1}{2}(28-22)f(22).$$

Here $f(8) = (8-4)/120 = \frac{1}{30}$ and $f(22) = (28-22)/168 = \frac{1}{28}$ min^{-1}.

$$\begin{aligned} P &= 1 - \tfrac{1}{2}(4)(\frac{1}{30}) - \tfrac{1}{2}(6)(\frac{1}{28}) \\ &= 1 - \frac{1}{15} - \frac{3}{28} = \boxed{\frac{347}{420} \approx 0.8262 = 82.62\%}. \end{aligned}$$

Shaded event: $8 \le S \le 22$; s in minutes



Density has units min^{-1}. Integrating over $[8, 14]$ and $[14, 22]$ gives the same probability.

Return to this task

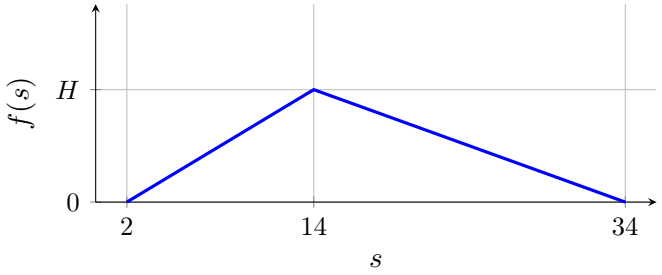
Solution: Changed range

What stays fixed. The mode is 14 min, the event is $[8, 22]$, and both branches remain straight. Only the support endpoints change.

Stage 1: Support and shape

$S \sim \text{Tri}(2, 34, 14)$ has support $[2, 34]$ min. The density rises from $(2, 0)$ to $(14, H)$ and falls to $(34, 0)$.

Support and triangular shape; horizontal coordinate in minutes



Stage 2: Total area = 1

The triangle area is $\frac{1}{2} \times$ base $\times$ height, so

$$1 = \frac{1}{2}(34-2)H \quad \Longrightarrow \quad H = \frac{2}{32} = \frac{1}{16}\ \text{min}^{-1}.$$

Stage 3: Complete piecewise pdf

Use $m = (y_2 - y_1)/(x_2 - x_1)$ and $q = y_1 - mx_1$ in $f(s) = ms + q$. For the rising line through $(2, 0)$ and $(14, \frac{1}{16})$,

$$m_1 = \frac{\frac{1}{16} - 0}{14 - 2} = \frac{1}{192}, \qquad q_1 = 0 - (\frac{1}{192})(2) = \frac{-1}{96}, \qquad f_1(s) = \frac{s-2}{192}.$$

For the falling line through $(14, \frac{1}{16})$ and $(34, 0)$,

$$m_2 = \frac{0 - \frac{1}{16}}{34 - 14} = \frac{-1}{320}, \qquad q_2 = 0 - (\frac{-1}{320})(34) = \frac{17}{160}.$$

Thus the complete density is

$$f(s) = \begin{cases} \dfrac{s-2}{192}, & 2 \le s \le 14, \\ \dfrac{34-s}{320}, & 14 < s \le 34, \\ 0, & \text{otherwise.} \end{cases}$$

The branches meet at height $\frac{1}{16}$, are nonnegative, and enclose total area 1.

Stage 4: Probability by complementary tail triangles

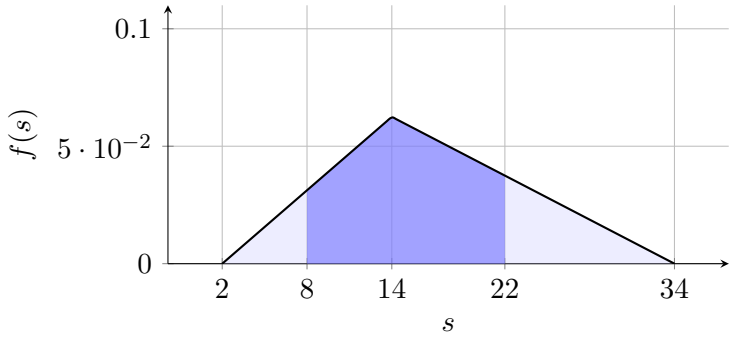
The interval straddles the mode; both tails must be removed:

$$P(8 \le S \le 22) = 1 - \frac{1}{2}(8-2)f(8) - \frac{1}{2}(34-22)f(22).$$

Here $f(8) = (8-2)/192 = \frac{1}{32}$ and $f(22) = (34-22)/320 = \frac{3}{80}$ min^{-1}.

$$\begin{aligned} P &= 1 - \tfrac{1}{2}(6)(\frac{1}{32}) - \tfrac{1}{2}(12)(\frac{3}{80}) \\ &= 1 - \frac{3}{32} - \frac{9}{40} = \boxed{\frac{109}{160} \approx 0.6813 = 68.13\%}. \end{aligned}$$

Shaded event: $8 \le S \le 22$**;** s **in minutes**



Density has units min^{-1}. Integrating over $[8, 14]$ and $[14, 22]$ gives the same probability.

Return to this task

3 A repeating irrigation cycle [C3-3]

A flower farm's pressure fluctuation occurs uniformly within a repeating 48-minute irrigation cycle, starting at minute 0 and restarting immediately after minute 48.

Questions/Tasks

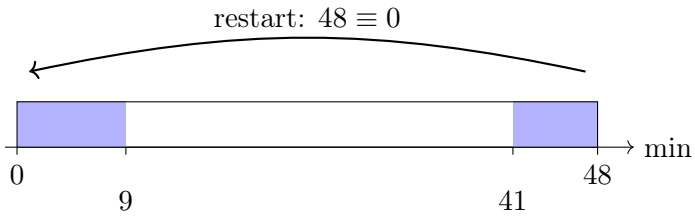
C3 Find the probability of a fluctuation from minute 41 through the restart to minute 9 of the next cycle. State the event as interval(s) within one cycle. **[1]**

C3 Find the probability of a fluctuation within 150 seconds on either side of the cycle start. State the event as interval(s) within one cycle. **[2]**

Solution: Window through the restart

Let $U \sim U(0, 48)$ represent position in minutes, with density $1/48$ min^{-1} on $[0, 48)$. The window splits into $[41, 48)$ and $[0, 9]$, of lengths 7 and 9 minutes.

The shaded lengths meet through the restart



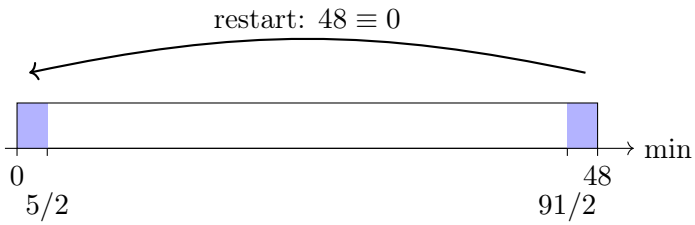
$$P = \frac{(48 - 41) + (9 - 0)}{48} = \frac{1}{3} \approx 0.3333 = 33.33\% .$$

Return to this task

Solution: Tolerance on both sides

Convert first: 150 s $= 150/60 = 5/2$ min on *each* side. The intervals are $[0, 5/2]$ and $[48 - 5/2, 48) = [91/2, 48)$.

Two equal pieces beside the cycle start



$$P = \frac{5/2 + 5/2}{48} = \frac{5}{48} \approx 0.1042 = 10.42\% .$$

Endpoints have probability zero; their inclusion does not change either answer.

Return to this task

4 Direction of a floating current marker [C3-3]

A river-monitoring team's floating marker has an initial direction uniform around a full circle. A bank marker indicates north. Angular tolerances apply on **both sides** of north, including wraparound.

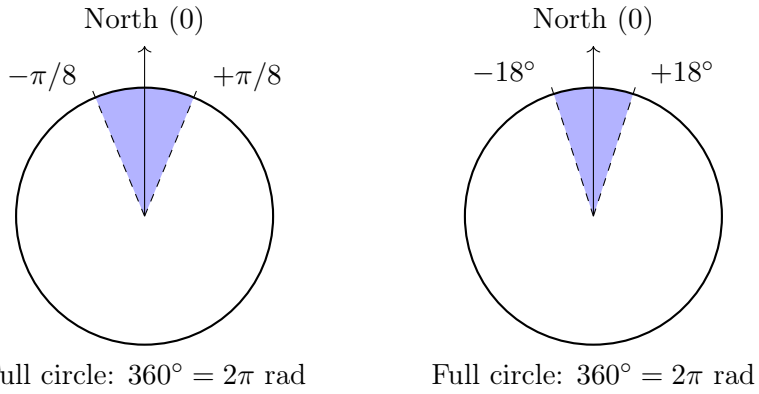
Questions/Tasks

C3 For each tolerance ($\pi/8$ radians and $18°$ from north), state the event as angular intervals with north at zero and calculate its probability. Identify the larger probability, give the difference in percentage points, and calculate the relative increase from the $18°$ probability. **[3]**

Solution: Angular probabilities and comparison

Uniform directions give probability equal to favourable angle divided by full-circle angle. Count *both sides* of north.

Two wraparound sectors centred at north



In radians, the event is $[0, \pi/8] \cup [2\pi - \pi/8, 2\pi)$:

$$P_{\pi/8} = \frac{2(\pi/8)}{2\pi} = \boxed{\frac{1}{8} = 0.1250 = 12.50\%}\,.$$

In degrees, the event is $[0, 18] \cup [342, 360)$:

$$P_{18°} = \frac{2(18°)}{360°} = \boxed{\frac{1}{10} = 0.1000 = 10.00\%}\,.$$

The $\pi/8$ tolerance (22.5 degrees on each side) gives greater probability:

$$\Delta P = \frac{1}{8} - \frac{1}{10} = \boxed{\frac{1}{40} = 0.0250}\,,$$

an absolute difference of **2.50 percentage points**. Using the 18-degree probability as baseline, the relative increase is

$$100\frac{1/40}{1/10}\% = 25.00\%.$$

Return to this task

5 Public-library return arrivals [C4-3]

Returns enter a public library's automated chute as a Poisson process, with a different constant rate in each scenario below. Let T be the waiting time in minutes from a fixed observation time to the next return.

For each scenario, identify the waiting-time model, rate in returns per minute and support, and calculate the probability. Give the general density function and briefly justify the model.

Questions/Tasks

C4 With an average of 7 returns every 35 minutes, find $P(T < 3)$. **[0.5]**

C4 With a mean waiting time of 5 minutes between returns, find $P(T > 8)$. **[0.5]**

C4 With an average of 4 returns every 25 minutes, find $P(1 < T < 6)$. **[1]**

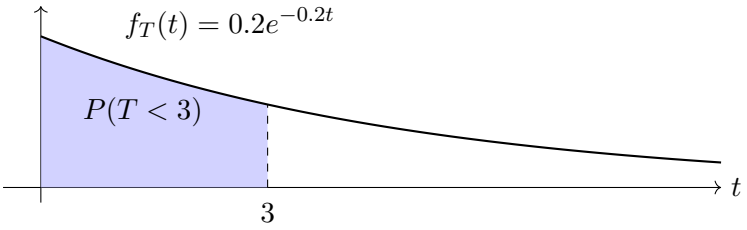
C4 With an average of 18 returns per hour, find $P(T < 2 \text{ or } T > 7)$. **[1]**

Solution: Less than three minutes

The rate is $\lambda_1 = 7/35 = 1/5 = 0.200$ returns/min. For $T \sim \text{Exp}(\lambda_1)$, $P(T < t) = 1 - e^{-\lambda_1 t}$.

$$P(T < 3) = 1 - e^{-(1/5)(3)} = 1 - e^{-3/5} \approx 0.4512 = 45.12\%.$$

Shaded event; t in minutes and $f_T(t)$ in inverse minutes



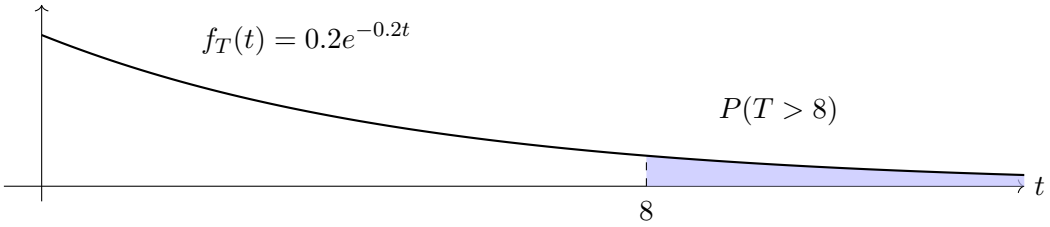
Return to this task

Solution: More than eight minutes

A mean wait of 5 min gives $\lambda_2 = 1/E(T) = 1/5 = 0.200$ returns/min. Use the survival formula $P(T > t) = e^{-\lambda_2 t}$.

$$P(T > 8) = e^{-(1/5)(8)} = \boxed{e^{-8/5} \approx 0.2019 = 20.19\%}.$$

Shaded event; t in minutes and $f_T(t)$ in inverse minutes



The shaded right tail continues beyond the plotted edge to infinity.

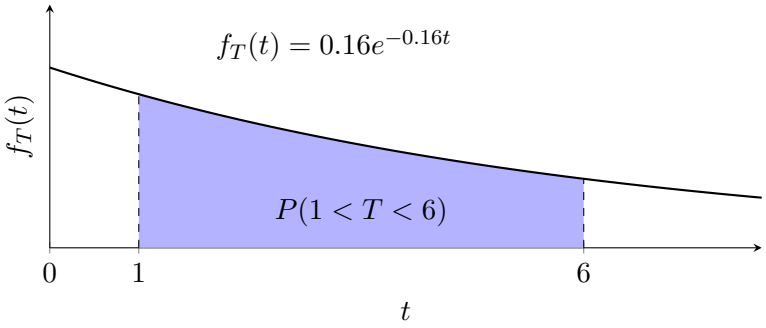
Return to this task

Solution: Between one and six minutes

The separate rate is $\lambda_3 = 4/25 = 0.160$ returns/min. For an interval, $P(a < T < b) = e^{-\lambda_3 a} - e^{-\lambda_3 b}$.

$$P(1 < T < 6) = e^{-(4/25)(1)} - e^{-(4/25)(6)} = \boxed{e^{-4/25} - e^{-24/25} \approx 0.4693 = 46.93\%}.$$

Shaded event; t in minutes and $f_T(t)$ in inverse minutes

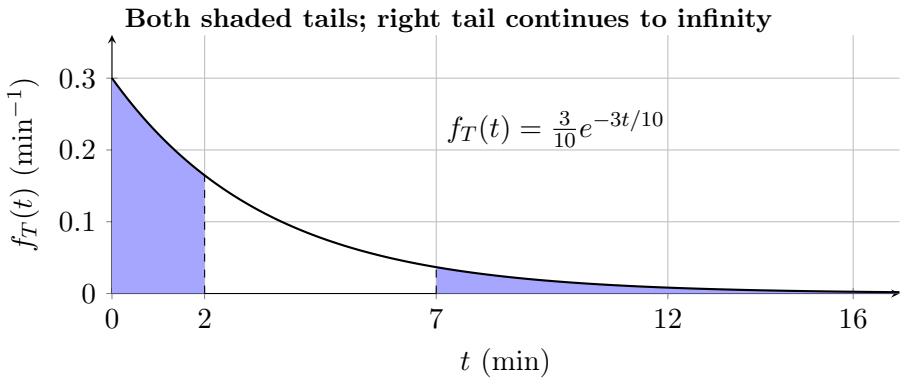


Return to this task

Solution: Outside two to seven minutes

Convert the hourly rate: $\lambda_4 = 18/60 = 3/10 = 0.300$ returns/min. The two tails are disjoint, so $P(T < a \text{ or } T > b) = (1 - e^{-\lambda_4 a}) + e^{-\lambda_4 b}$.

$$P(T < 2 \text{ or } T > 7) = 1 - e^{-(3/10)(2)} + e^{-(3/10)(7)} = \boxed{1 - e^{-3/5} + e^{-21/10} \approx 0.5736 = 57.36\%}.$$



For all four scenarios, $f_T(t) = \lambda e^{-\lambda t}$ for $t \geq 0$, zero otherwise, with support $[0, \infty)$. The constant-rate Poisson assumption makes exponential waiting times appropriate.

Return to this task

6 Bicycle-hire arrival rates [C4-2]

At a municipal bicycle-hire kiosk, rentals start as a Poisson process with different constant morning and afternoon rates. Let T_m, T_a be the respective waiting times in minutes from a fixed time in each shift to the next rental. Assume each shift continues throughout the wait.

Questions/Tasks

C4 In the morning, $P(T_m > 12) = 0.35$. Infer the rate in rental starts per minute and per hour. **[1]**

C4 In the afternoon, $P(T_a < 9) = 0.72$. Infer the rate in rental starts per minute and per hour. **[1]**

Solution: Morning rate from survival

Use $P(T_m > t) = e^{-\lambda_m t}$:

$$0.35 = e^{-12\lambda_m} \implies \ln(0.35) = -12\lambda_m \implies \lambda_m = -\frac{\ln(0.35)}{12} \approx 0.087 \text{ starts/min}.$$

Convert the exact rate to hourly units:

$$60\lambda_m = -5\ln(0.35) \approx \boxed{5.249 \text{ starts/hour}}.$$

Return to this task

Solution: Afternoon rate from a cumulative probability

Use $P(T_a < t) = 1 - e^{-\lambda_a t}$:

$$0.72 = 1 - e^{-9\lambda_a} \implies e^{-9\lambda_a} = 0.28 \implies -9\lambda_a = \ln(0.28).$$

$$\lambda_a = -\frac{\ln(0.28)}{9} \approx 0.141 \text{ starts/min}, \qquad 60\lambda_a = -\frac{20}{3}\ln(0.28) \approx \boxed{8.486 \text{ starts/hour}}.$$

Return to this task

7 Two recycling collection gates [C4-2]

Vehicle entries at recycling gates A and B follow independent constant-rate Poisson processes. From a fixed observation time, A's next entry has probability 0.84 of occurring within 4 minutes; B's has probability 0.60 of taking more than 4 minutes. Count both gates' entries from this same time.

Questions/Tasks

C4 Infer both rates in entries per minute and state the expected-count functions. Find when A's expected count exceeds B's by 4, in minutes, and explain what this time means for actual entries. **[2]**

Solution: Rates and expected-count difference

For A the given probability is cumulative:

$$0.84 = 1 - e^{-4\lambda_A} \implies e^{-4\lambda_A} = 0.16 \implies \lambda_A = -\frac{\ln(0.16)}{4} \approx 0.458 \text{ entries/min}.$$

For B it is a survival probability:

$$0.60 = e^{-4\lambda_B} \implies \lambda_B = -\frac{\ln(0.60)}{4} \approx 0.128 \text{ entries/min}.$$

Define $N_A(t), N_B(t)$ as entry counts since the common observation time. For constant-rate Poisson processes,

$$E[N_A(t)] = \lambda_A t, \qquad E[N_B(t)] = \lambda_B t.$$

Thus $E[N_A(t)] - E[N_B(t)] = (\lambda_A - \lambda_B)t$. Since $\lambda_A > \lambda_B$, the required time is positive and finite:

$$t = \frac{4}{\lambda_A - \lambda_B} = \frac{16}{\ln(0.60/0.16)} = \boxed{\frac{16}{\ln(15/4)} \approx 12.105 \text{ min}}.$$

This specifies a difference of 4 in *expected* counts; the actual random count difference need not be 4 at this time.

Return to this task

8 Two document-digitisation desks [C5-2]

At a municipal archive, A and B are file-digitisation times in minutes at two desks, both with support $[2, 17]$. A's density rises linearly from zero at 2 to its maximum at 17. B has equal probabilities for equal-length intervals within the support. Both densities are zero elsewhere.

Questions/Tasks

C5 Identify A's distribution, parameters and support, and calculate its mean, variance and standard deviation. Interpret the mean in context. **[1]**

C5 Identify B's distribution, parameters and support, and calculate its mean, variance and standard deviation. Interpret the mean and standard deviation in context. **[1]**

Solution: Increasing-linear digitisation time

The increasing straight density is an endpoint triangular model: $A \sim \text{Tri}(2, 17, 17)$, support $[2, 17]$ min. For increasing-linear density on $[a, b]$,

$$\mu = \frac{a + 2b}{3}, \qquad \sigma^2 = \frac{(b-a)^2}{18}, \qquad \sigma = \sqrt{\sigma^2}.$$

$$E(A) = \frac{2 + 2(17)}{3} = 12 \text{ min}, \qquad \text{Var}(A) = \frac{(17-2)^2}{18} = \frac{25}{2} = 12.500 \text{ min}^2,$$

$$\sigma_A = \sqrt{25/2} \approx 3.536 \text{ min}.$$

Over many files, desk A averages 12 minutes per file.

Return to this task

Solution: Uniform digitisation time

Equal-length intervals have equal probabilities, so $B \sim U(2, 17)$, support $[2, 17]$ min. The formulas are $\mu = (a+b)/2$, $\sigma^2 = (b-a)^2/12$, $\sigma = \sqrt{\sigma^2}$.

$$E(B) = \frac{2 + 17}{2} = 9.500 \text{ min}, \qquad \text{Var}(B) = \frac{(17-2)^2}{12} = \frac{75}{4} = 18.750 \text{ min}^2,$$

$$\sigma_B = \sqrt{75/4} \approx 4.330 \text{ min}.$$

Desk B averages 9.5 minutes per file; its time SD is about 4.330 minutes.

Return to this task

9 Pump maintenance and daily operating costs [C5-2]

A rural water association's pump maintenance duration D has support $[2, 14]$ hours. Its density falls linearly from its maximum at 2 to zero at 14 and is zero elsewhere.

Daily operating cost R, in **thousands of Colombian pesos (COP)**, has an exact symmetric bell-shaped density on the real line, centred at 2400 with inflection points 2250 and 2550.

Questions/Tasks

C5 Identify D's distribution, parameters and support, and calculate its mean, variance and standard deviation. Interpret the mean in context. **[1]**

C5 Identify R's distribution and infer its parameters and support. Calculate mean and SD in **COP** and variance in **COP**^{2}. Interpret the mean. **[1]**

Solution: Decreasing-linear maintenance duration

The decreasing straight density gives $D \sim \text{Tri}(2, 14, 2)$, support $[2, 14]$ h. Use $\mu = (2a + b)/3$, $\sigma^2 = (b - a)^2/18$, $\sigma = \sqrt{\sigma^2}$:

$$E(D) = \frac{2(2) + 14}{3} = 6 \text{ h}, \qquad \text{Var}(D) = \frac{(14 - 2)^2}{18} = 8 \text{ h}^2, \qquad \sigma_D = \sqrt{8} \approx 2.828 \text{ h}.$$

The model predicts a long-run average maintenance duration of 6 hours.

Return to this task

Solution: Normal cost and conversion to COP

The exact symmetric bell curve identifies a normal distribution. In thousands of COP, the centre is $\mu_R = 2400$. Normal inflection points are $\mu_R \pm \sigma_R$, so

$$\sigma_R = 2550 - 2400 = 2400 - 2250 = 150, \qquad R \sim N(2400, 150^2) = N(2400, 22500).$$

Its mathematical support is $\mathbb{R}$. In these units, mean is μ_R, variance is σ_R^2, and SD is σ_R. Let $Y = 1000R$ be cost in COP. The conversion laws are

$$E(cR) = cE(R), \quad \text{Var}(cR) = c^2 \,\text{Var}(R), \quad \text{SD}(cR) = |c| \,\text{SD}(R).$$

Therefore

$$E(Y) = 1000(2400) = 2{,}400{,}000 \text{ COP},$$

$$\text{Var}(Y) = 1000^2(22500) = 22{,}500{,}000{,}000 \text{ COP}^2, \qquad \text{SD}(Y) = 1000(150) = 150{,}000 \text{ COP}.$$

The association's long-run average daily operating cost is 2,400,000 COP; individual days vary around it.

Return to this task

10 Museum admissions and audio-guide sessions [C5-2]

Museum admissions follow a constant-rate Poisson process averaging 15 per hour. Let T be the interarrival time in minutes.

Audio-guide session duration S has support $[6, 27]$ minutes. Its density rises linearly from zero at 6 to a single peak at 15, then falls linearly to zero at 27; it is zero elsewhere.

Questions/Tasks

C5 Identify T's distribution, parameters and support, and calculate its mean and SD in minutes and its variance in minutes^2. Interpret the mean in context. **[1]**

C5 Identify S's distribution, parameters and support, and calculate its mean and SD in minutes and its variance in minutes^2. Interpret the mean and standard deviation in context. **[1]**

Solution: Exponential interarrival time

The constant-rate Poisson process gives $T \sim \text{Exp}(\lambda)$, support $[0, \infty)$ min, with $\lambda = 15/60 = 1/4 = 0.250$ admissions/min. Use $\mu = 1/\lambda$, $\sigma^2 = 1/\lambda^2$, $\sigma = 1/\lambda$:

$$E(T) = \frac{1}{1/4} = 4 \text{ min}, \qquad \text{Var}(T) = \frac{1}{(1/4)^2} = 16 \text{ min}^2, \qquad \sigma_T = 4 \text{ min}.$$

The mean gap between admissions is 4 minutes.

Return to this task

Solution: Triangular audio-guide duration

The straight branches give $S \sim \text{Tri}(6, 27, 15)$, support $[6, 27]$ min. For minimum a, maximum b, mode c,

$$\mu = \frac{a+b+c}{3}, \quad \sigma^2 = \frac{a^2+b^2+c^2-ab-ac-bc}{18}, \quad \sigma = \sqrt{\sigma^2}.$$

$$E(S) = \frac{6+27+15}{3} = 16 \text{ min}.$$

$$\text{Var}(S) = \frac{6^2+27^2+15^2-6(27)-6(15)-27(15)}{18}$$

$$= \frac{333}{18} = \frac{37}{2} = 18.500 \text{ min}^2,$$

$$\sigma_S = \sqrt{37/2} \approx 4.301 \text{ min}.$$

The model predicts a mean session of 16 minutes with SD about 4.301 minutes.

Return to this task

11 Commercial-kitchen water use [C6-2]

A catering kitchen records water use X in litres for 50 cleaning batches in two non-overlapping parts. The first 20 observations are:

4.1	4.5	5.0	5.3	5.8
6.0	6.2	6.7	7.0	7.4
4.8	5.0	5.5	5.9	6.4
6.8	7.0	7.2	7.6	8.0

The remaining 30 observations have this discrete frequency table:

x_i	5.0	5.4	5.8	6.3	7.0	7.5
f_i	4	6	5	7	3	5

Combine both parts as one sample. The model gives $P(5 \le X < 7) = 0.58$.

Questions/Tasks

C6 State the sample size and event count, then calculate the empirical probability. **[1]**

C6 Calculate its absolute and percentage errors. **[1]**

Solution: Empirical probability

Eligible raw observations are $5.0, 5.3, 5.8, 6.0, 6.2, 6.7, 5.0, 5.5, 5.9, 6.4, 6.8$, giving **11** batches. Table values $5.0, 5.4, 5.8, 6.3$ contribute $4 + 6 + 5 + 7 = 22$ batches. Include every 5 and exclude every 7. The records do not overlap, so $n = 20 + 30 = 50$ and $k = 11 + 22 = 33$.

$$\widehat{P}(5 \le X < 7) = \frac{33}{50} = 0.6600 = 66.00\%.$$

Return to this task

Solution: Absolute and percentage errors

Use $E = |\widehat{P} - P_0|$ and $E\% = 100E/|P_0|$, using the exact relative frequency:

$$E = \left|\frac{33}{50} - 0.58\right| = \frac{2}{25} \approx 0.0800.$$

$$E\% = 100\frac{\frac{2}{25}}{0.58}\% \approx 13.79\%.$$

Probabilities and their errors are unitless.

Return to this task

12 Produce cooling, grouped data [C6-2]

Produce-crate cooling times X in minutes at a packing centre are grouped below. Classes include the left endpoint and exclude the right.

Class	$[2,7)$	$[7,12)$	$[12,17)$	$[17,22)$
f_i	4	7	6	4

The model gives $P(7 \le X < 17) = 0.68$.

Questions/Tasks

C6 State the sample size and event count, then calculate the empirical probability. **[1]**

C6 Calculate its absolute and percentage errors. **[1]**

Solution: Empirical probability

The complete included classes are $[7,12)$ and $[12,17)$. Thus $n = 4+7+6+4 = 21$ and $k = 7+6 = 13$. These counts are exact because event endpoints are class boundaries; no midpoint approximation is needed.

$$\widehat{P}(7 \le X < 17) = \frac{13}{21} \approx 0.6190 = 61.90\%.$$

Return to this task

Solution: Absolute and percentage errors

Use $E = |\widehat{P} - P_0|$ and $E\% = 100E/|P_0|$, using the exact relative frequency:

$$E = \left|\frac{13}{21} - 0.68\right| = \frac{32}{525} \approx 0.0610.$$

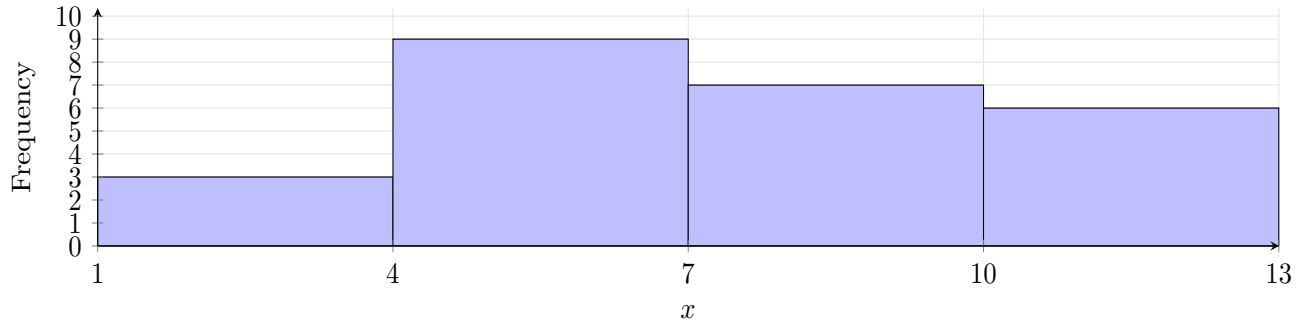
$$E\% = 100\frac{\frac{32}{525}}{0.68}\% \approx 8.96\%.$$

Probabilities and their errors are unitless.

Return to this task

13 Battery-swap service, histogram [C6-2]

The histogram is the only sample record of battery-swap times X at an electric-motorcycle station. The horizontal axis shows minutes; bar heights are frequencies. Equal-width classes include the left endpoint and exclude the right.



The model gives $P(4 \le X < 10) = 0.60$.

Questions/Tasks

C6 State the sample size and event count, then calculate the empirical probability. **[1]**

C6 Calculate its absolute and percentage errors. **[1]**

Solution: Empirical probability

Read the four histogram heights as frequencies $3, 9, 7, 6$. The included complete classes are $[4, 7)$ and $[7, 10)$, giving $n = 3 + 9 + 7 + 6 = 25$ and $k = 9 + 7 = 16$. No midpoint approximation is needed.

$$\widehat{P}(4 \le X < 10) = \frac{16}{25} = 0.6400 = 64.00\%.$$

Return to this task

Solution: Absolute and percentage errors

Use $E = |\widehat{P} - P_0|$ and $E\% = 100E/|P_0|$, using the exact relative frequency:

$$E = \left|\frac{16}{25} - 0.60\right| = \frac{1}{25} \approx 0.0400.$$

$$E\% = 100\frac{\frac{1}{25}}{0.60}\% \approx 6.67\%.$$

Probabilities and their errors are unitless.

Return to this task

14 Daily paper use, mixed data [C7-2]

A legal-aid office records paper use in hundreds of sheets on 13 days in two non-overlapping parts: the first five observations are $13, 16, 18, 21, 23$; the remaining eight are summarised below:

x_i	13	18	23
f_i	1	4	3

Combine both parts as one sample. The theoretical mean is 18 hundreds of sheets and variance is 16 (hundreds of sheets)^2.

Questions/Tasks

C7 State the sample size; calculate the sample mean, variance and standard deviation. Show your frequency summary and weighted sum of squared deviations. **[1]**

C7 Calculate absolute and percentage errors for mean and variance. **[1]**

Solution: Sample statistics from combined data

The five raw observations contribute one each at $13, 16, 18, 21, 23$; the table adds $1, 4, 3$ at $13, 18, 23$. The combined actual-value frequencies below require no midpoint estimation.

x_i	13	16	18	21	23
f_i	2	1	5	1	4

Use sample variance with denominator $n - 1$:

$$n = \sum f_i = 13, \qquad \bar{x} = \frac{\sum f_i x_i}{n}.$$

$$\bar{x} = \frac{2(13) + 1(16) + 5(18) + 1(21) + 4(23)}{13} = \frac{245}{13} \approx 18.846 \text{ hundreds of sheets.}$$

The weighted squared-deviation sum is

$$\begin{aligned} Q &= \sum_i f_i (x_i - \bar{x})^2 \\ &= 2\left(13 - \frac{245}{13}\right)^2 + 1\left(16 - \frac{245}{13}\right)^2 + 5\left(18 - \frac{245}{13}\right)^2 \\ &\quad + 1\left(21 - \frac{245}{13}\right)^2 + 4\left(23 - \frac{245}{13}\right)^2 \\ &= \frac{1998}{13} \text{ (hundreds of sheets)}^2. \end{aligned}$$

$$s^2 = \frac{\sum_i f_i (x_i - \bar{x})^2}{n - 1} = \frac{\frac{1998}{13}}{13 - 1} = \frac{333}{26} \approx 12.808 \text{ (hundreds of sheets)}^2,$$

$$s = \sqrt{s^2} \approx \sqrt{\frac{333}{26}} \approx 3.579 \text{ hundreds of sheets.}$$

Return to this task

Solution: Mean and variance errors

For a statistic z and theoretical value z_0, use $E = |z - z_0|$, $E\% = 100E/|z_0|$. Using unrounded statistics:

$$E_{\bar{x}} = \left|\frac{245}{13} - 18\right| = \frac{11}{13} \approx 0.846 \text{ hundreds of sheets}, \qquad E_{\bar{x}}\% = 100\frac{\frac{11}{13}}{18}\% \approx 4.70\%.$$

$$E_{s^2} = \left|\frac{333}{26} - 16\right| = \frac{83}{26} \approx 3.192 \text{ (hundreds of sheets)}^2, \qquad E_{s^2}\% = 100\frac{\frac{83}{26}}{16}\% \approx 19.95\%.$$

The requested errors concern the mean and variance only.

Return to this task

15 Compost maturation checks, grouped data [C7-2]

An urban garden groups times between compost checks in days. Classes include the left endpoint and exclude the right. The theoretical mean is 15 days and variance is 24 days^2.

Class	$[5, 11)$	$[11, 17)$	$[17, 23)$
f_i	3	2	2

Questions/Tasks

C7 State the sample size; calculate the sample mean, variance and standard deviation. Show your frequency summary and weighted sum of squared deviations. **[1]**

C7 Calculate absolute and percentage errors for mean and variance. **[1]**

Solution: Sample statistics from class midpoints

The class midpoint is (left + right)/2. Individual observations are unavailable, so all three statistics and their errors are **midpoint estimates**.

m_i	8	14	20
f_i	3	2	2

Use sample variance with denominator $n-1$:

$$n = \sum f_i = 7, \qquad \bar{x} = \frac{\sum f_i m_i}{n}.$$

$$\bar{x} \approx \frac{3(8) + 2(14) + 2(20)}{7} = \frac{92}{7} \approx 13.143 \text{ days}.$$

The weighted squared-deviation sum is

$$\begin{aligned} Q &= \sum_i f_i (m_i - \bar{x})^2 \\ &= 3\left(8 - \frac{92}{7}\right)^2 + 2\left(14 - \frac{92}{7}\right)^2 + 2\left(20 - \frac{92}{7}\right)^2 \\ &= \frac{1224}{7} \text{ (days)}^2. \end{aligned}$$

$$s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1} \approx \frac{\frac{1224}{7}}{7-1} = \frac{204}{7} \approx 29.143 \text{ (days)}^2,$$

$$s = \sqrt{s^2} \approx \sqrt{\frac{204}{7}} \approx 5.398 \text{ days}.$$

Return to this task

Solution: Mean and variance errors

For a statistic z and theoretical value z_0, use $E = |z - z_0|$, $E\% = 100E/|z_0|$. Using unrounded midpoint estimates:

$$E_{\bar{x}} = \left|\frac{92}{7} - 15\right| = \frac{13}{7} \approx 1.857 \text{ days}, \qquad E_{\bar{x}}\% = 100\frac{\frac{13}{7}}{15}\% \approx 12.38\%.$$

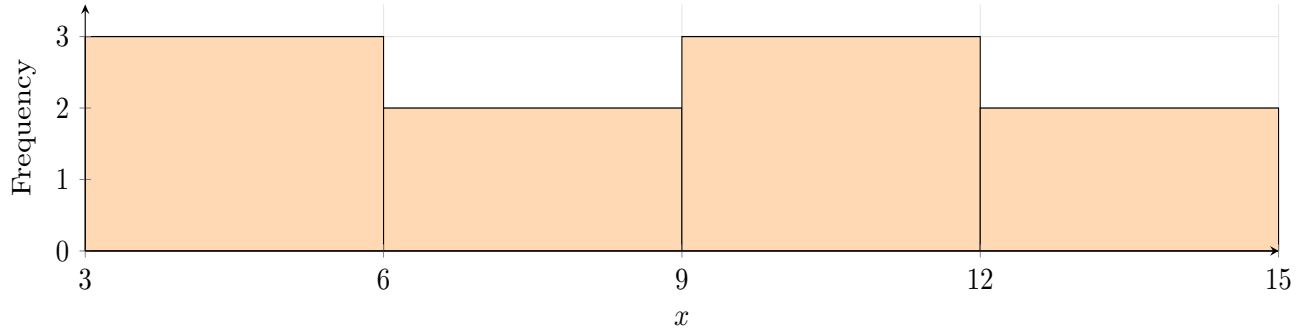
$$E_{s^2} = \left|\frac{204}{7} - 24\right| = \frac{36}{7} \approx 5.143 \text{ (days)}^2, \qquad E_{s^2}\% = 100\frac{\frac{36}{7}}{24}\% \approx 21.43\%.$$

The requested errors concern the mean and variance only.

Return to this task

16 Community-bakery flour use, histogram [C7-2]

The histogram is the only sample record of flour use per batch at a community bakery. The horizontal axis shows kilograms; bar heights are frequencies. Equal-width classes include the left endpoint and exclude the right. The theoretical mean is 8 kg and variance is 7 kg^2.



Questions/Tasks

C7 State the sample size; calculate the sample mean, variance and standard deviation. Show your frequency summary and weighted sum of squared deviations. **[1]**

C7 Calculate absolute and percentage errors for mean and variance. **[1]**

Solution: Sample statistics from class midpoints

The class midpoint is (left + right)/2. Individual observations are unavailable, so all three statistics and their errors are **midpoint estimates**.

m_i	$\frac{9}{2}$	$\frac{15}{2}$	$\frac{21}{2}$	$\frac{27}{2}$
f_i	3	2	3	2

Use sample variance with denominator $n-1$:

$$n = \sum f_i = 10, \qquad \bar{x} = \frac{\sum f_i m_i}{n}.$$

$$\bar{x} \approx \frac{3(\frac{9}{2}) + 2(\frac{15}{2}) + 3(\frac{21}{2}) + 2(\frac{27}{2})}{10} = \frac{87}{10} \approx 8.700 \text{ kg}.$$

The weighted squared-deviation sum is

$$\begin{aligned} Q &= \sum_i f_i (m_i - \bar{x})^2 \\ &= 3\left(\frac{9}{2} - \frac{87}{10}\right)^2 + 2\left(\frac{15}{2} - \frac{87}{10}\right)^2 + 3\left(\frac{21}{2} - \frac{87}{10}\right)^2 \\ &\quad + 2\left(\frac{27}{2} - \frac{87}{10}\right)^2 \\ &= \frac{558}{5} \text{ (kg)}^2. \end{aligned}$$

$$s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1} \approx \frac{\frac{558}{5}}{10-1} = \frac{62}{5} \approx 12.400 \text{ (kg)}^2,$$

$$s = \sqrt{s^2} \approx \sqrt{\frac{62}{5}} \approx 3.521 \text{ kg}.$$

Return to this task

Solution: Mean and variance errors

For a statistic z and theoretical value z_0, use $E = |z - z_0|$, $E\% = 100E/|z_0|$. Using unrounded midpoint estimates:

$$E_{\bar{x}} = \left|\frac{87}{10} - 8\right| = \frac{7}{10} \approx 0.700 \text{ kg}, \qquad E_{\bar{x}}\% = 100\frac{\frac{7}{10}}{8}\% \approx 8.75\%.$$

$$E_{s^2} = \left|\frac{62}{5} - 7\right| = \frac{27}{5} \approx 5.400 \text{ (kg)}^2, \qquad E_{s^2}\% = 100\frac{\frac{27}{5}}{7}\% \approx 77.14\%.$$

The requested errors concern the mean and variance only.

Return to this task

17 Summarized Solutions

C1 – All twelve variables

Problem 1.

Context	Variable	Class	Model (D: discrete; C: continuous)
1	a	D	Bernoulli(0.86)
1	b	D	Binomial$(35, 0.86)$
2	a	D	Geometric(0.06), includes first faulty meter
2	b	C	$U(-2, 4)$ L
3	a	D	Poisson(3) bookings in one hour
3	b	C	Tri$(8, 26, 15)$ min
4	a	C	Increasing-linear: $200(x - 0.2)/49$ on $[0.2, 0.9]$
4	b	C	$N(1, 9)$ in watts
5	a	D	Binomial$(45, 0.07)$
5	b	D	Poisson(24) calls over four days
6	a	C	Tri$(12, 36, 12)$ min (assuming linearity)
6	b	C	$N(18, 6.25)$ in kg (assuming normality)

- Context 6: neither model is unique; steady decrease need not be linear, and symmetry need not be normal.
- Normal candidate: support $\mathbb{R}$; physical use is nonnegative, exact support unspecified.

C2 – Triangular models

Problem 2; mode 14 min retained.

Support (min)	Height (min^{-1})	$P(8 \le S \le 22)$
$[4, 28]$	$2/24 = 1/12$	$1 - 4^2/240 - 6^2/336 = 347/420 \approx 0.8262$
$[2, 34]$	$2/32 = 1/16$	$1 - 6^2/384 - 12^2/640 = 109/160 \approx 0.6813$

C3 – Cycles and directions

Problem	Event	Probability
3a	$[41, 48) \cup [0, 9]$ min	$((48 - 41) + 9)/48 = 1/3 \approx 0.3333$
3b	$[0, 5/2] \cup [91/2, 48)$ min	$2(150/60)/48 = 5/48 \approx 0.1042$
4	$\pi/8$ **rad:** $[0, \pi/8] \cup [15\pi/8, 2\pi)$	$\mathbf{2}(\pi/\mathbf{8})/(\mathbf{2}\pi) = \mathbf{1/8} = \mathbf{0.1250}$
	$[0, 18] \cup [342, 360)$ degrees	$2(18)/360 = 1/10 = 0.1000$

Difference (percentage points): $100(1/8 - 1/10) = \mathbf{2.50}$; relative increase: $100(1/8 - 1/10)/(1/10) = 25.00\%$.

C4 – Waiting probabilities and inferred rates

Problem 5: $T \sim \text{Exp}(\lambda)$, support $[0, \infty)$ min; $f(t) = \lambda e^{-\lambda t}$ (min^{-1}) for $t \geq 0$, zero otherwise (constant-rate Poisson arrivals).

λ (min^{-1})	Event (T: min)	Probability
$7/35 = 1/5$	$T < 3$	$1 - e^{-(1/5)3} \approx 0.4512$
$1/5$	$T > 8$	$e^{-(1/5)8} \approx 0.2019$
$4/25$	$1 < T < 6$	$e^{-(4/25)1} - e^{-(4/25)6} \approx 0.4693$
$18/60 = 3/10$	$T < 2$ or $T > 7$	$1 - e^{-(3/10)2} + e^{-(3/10)7} \approx 0.5736$

Problem 6:

Shift	Rate (starts/min)	Rate (starts/hour)
Morning	$-\ln(0.35)/12 \approx 0.087$	$-5\ln(0.35) \approx 5.249$
Afternoon	$-\ln(1-0.72)/9 \approx 0.141$	$-(20/3)\ln(0.28) \approx 8.486$

Problem 7:

Channel/gate	Rate (min^{-1})	$E[N(t)]$ (t: min)
A	$-\ln(1-0.84)/4 \approx 0.458$	$[-\ln(0.16)/4]t$
B	$-\ln(0.60)/4 \approx 0.128$	$[-\ln(0.60)/4]t$

Expected-count difference 4: $t = 4/[(-\ln 0.16 + \ln 0.60)/4] = 16/\ln(15/4) \approx 12.105$ min; actual counts remain random.

C5 – Theoretical moments

Model (A, B, T, S: min; D: h)	Mean	Variance (unit^2)	SD
8: $A \sim \text{Tri}(2, 17, 17)$	$(2 + 2(17))/3 = 12.000$	$(17-2)^2/18 = 12.500$	$\sqrt{25/2} \approx 3.536$
8: $B \sim U(2, 17)$	$(2 + 17)/2 = 9.500$	$(17-2)^2/12 = 18.750$	$\sqrt{75/4} \approx 4.330$
9: $D \sim \text{Tri}(2, 14, 2)$	$(2(2) + 14)/3 = 6.000$	$(14-2)^2/18 = 8.000$	$\sqrt{8} \approx 2.828$
10: $T \sim \text{Exp}(15/60)$	$1/(1/4) = 4.000$	$1/(1/4)^2 = 16.000$	$\sqrt{16} \approx 4.000$
10: $S \sim \text{Tri}(6, 27, 15)$	$(6 + 27 + 15)/3 = 16.000$	$333/18 = 18.500$	$\sqrt{37/2} \approx 4.301$

9: $R \sim N(2400, 22500)$, support $\mathbb{R}$	Calculation
SD (thousands of COP)	$2550 - 2400 = 150$
Mean (COP)	$1000(2400) = 2{,}400{,}000$
Variance (COP^2)	$1000^2(22500) = 22{,}500{,}000{,}000$
SD (COP)	$1000(150) = 150{,}000$

Long-run average daily operating cost: 2,400,000 COP.

- A, B: mean digitisation time per file; B's SD measures variation around its mean.
- D: mean maintenance duration; T: mean admission gap; S: mean session duration and spread.

C6 – Empirical probabilities

Problem/event	Counts, probability and errors
11: $5 \le X < 7$ (L)	$n = 20 + 30 = 50,\ k = 11 + (4 + 6 + 5 + 7) = 33;\quad \widehat{P} = 33/50 = 0.6600$
Absolute; percentage error	$\lvert 33/50 - 0.58\rvert \approx 0.0800;\quad 100(2/25)/0.58 \approx 13.79\%$
12: $7 \le X < 17$ (min)	$n = 4 + 7 + 6 + 4 = 21,\ k = 7 + 6 = 13;\quad \widehat{P} = 13/21 \approx 0.6190$
Absolute; percentage error	$\lvert 13/21 - 0.68\rvert \approx 0.0610;\quad 100(32/525)/0.68 \approx 8.96\%$
13: $4 \le X < 10$ (min)	$n = 3 + 9 + 7 + 6 = 25,\ k = 9 + 7 = 16;\quad \widehat{P} = 16/25 = 0.6400$
Absolute; percentage error	$\lvert 16/25 - 0.6\rvert \approx 0.0400;\quad 100(1/25)/0.6 \approx 6.67\%$

C7 – Sample statistics and requested errors

14	Units: hundreds of sheets; variance: (hundreds of sheets)^2
x_i; f_i	$(13, 16, 18, 21, 23);\ (2, 1, 5, 1, 4);\ n = 13$
Mean	$(2(13) + 1(16) + 5(18) + 1(21) + 4(23))/13 = 245/13 \approx 18.846$
$Q = \sum f_i(x_i - \bar{x})^2$	$4771 - 245^2/13 = 1998/13$
Sample variance; SD	$s^2 = (1998/13)/(13 - 1) \approx 12.808;\ s = \sqrt{333/26} \approx 3.579$
Mean error; error (%)	$\lvert 245/13 - 18\rvert \approx 0.846;\quad 100(11/13)/18 \approx 4.70\%$
Variance error; error (%)	$\lvert 333/26 - 16\rvert \approx 3.192;\quad 100(83/26)/16 \approx 19.95\%$

15 (midpoint estimates)	Units: days; variance: (days)^2
m_i; f_i	$(8, 14, 20);\ (3, 2, 2);\ n = 7$
Mean	$(3(8) + 2(14) + 2(20))/7 = 92/7 \approx 13.143$
$Q = \sum f_i(m_i - \bar{x})^2$	$1384 - 92^2/7 = 1224/7$
Sample variance; SD	$s^2 = (1224/7)/(7 - 1) \approx 29.143;\ s = \sqrt{204/7} \approx 5.398$
Mean error; error (%)	$\lvert 92/7 - 15\rvert \approx 1.857;\quad 100(13/7)/15 \approx 12.38\%$
Variance error; error (%)	$\lvert 204/7 - 24\rvert \approx 5.143;\quad 100(36/7)/24 \approx 21.43\%$

16 (midpoint estimates)	Units: kg; variance: (kg)^2
m_i; f_i	$(4.5, 7.5, 10.5, 13.5);\ (3, 2, 3, 2);\ n = 10$
Mean	$(3(4.5) + 2(7.5) + 3(10.5) + 2(13.5))/10 = 87/10 \approx 8.700$
$Q = \sum f_i(m_i - \bar{x})^2$	$1737/2 - 87^2/10 = 558/5$
Sample variance; SD	$s^2 = (558/5)/(10 - 1) \approx 12.400;\ s = \sqrt{62/5} \approx 3.521$
Mean error; error (%)	$\lvert 87/10 - 8\rvert \approx 0.700;\quad 100(7/10)/8 \approx 8.75\%$
Variance error; error (%)	$\lvert 62/5 - 7\rvert \approx 5.400;\quad 100(27/5)/7 \approx 77.14\%$